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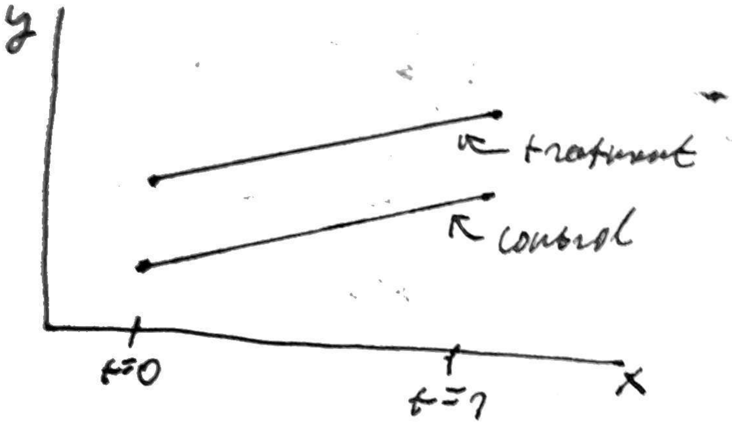
1. a The parallel trend assumption states that a certain trend is the same in both the untreated group and treated group. For example, using the model $y_{it} = B_0 + B_1 T_t + B_2 D_i + B_3 T_t \cdot D_i + u_{it}$ assumes that the trend for $t=0$ and $t=1$ is the same.

Graphically:



In such a case, as there is no deviation from the linear trend, we do not see a causal effect. Crucially for the parallel trend assumption, it assumes a constant difference between the two groups.

b. $E(y_{it} | T_t = 0, D_i = 0) = B_0$

$E(y_{it} | T_t = 1, D_i = 0) = B_0 + B_2 \cdot \underbrace{T_t}_{=1}$ (normally)

However, we have:

$E(y_{it} | T_t = 1, D_i = 0) = B_0 + (B_2 + \gamma) \underbrace{T_t}_{=1}$

We are interested in $E(\Delta y_{it}\ did) = E(\Delta y_{it}\ t=1) - E(\Delta y_{it}\ t=0)$

Graphically:

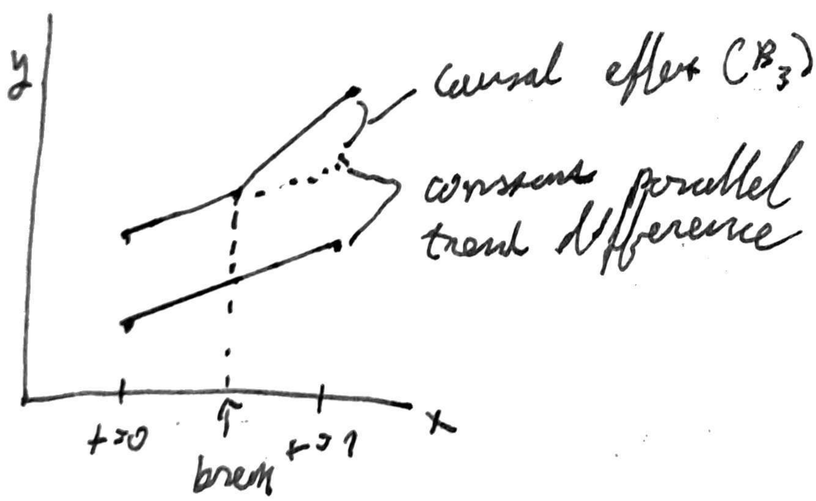
	$E(y_{it} \mid T_t = 0)$	$E(y_{it} \mid T_t = 1)$	difference
$E(y_{it} \mid D_i = 0)$	B_0	$B_0 + B_2 + \gamma$	$B_1 + \gamma$
$E(y_{it} \mid D_i = 1)$	$B_0 + B_2$	$B_0 + B_2 + B_1 + \cancel{\gamma} + B_3$	$B_1 + \cancel{\gamma} + B_3$
Difference	B_2	$B_2 + B_3 - \gamma$	$B_3 - \gamma$ ← bias

γ omitted because $D_i = 1$

effect of policy change

As a result we obtain a biased estimator of the policy change, i.e. $B_3 - \gamma$. Moreover, this is due to us calculating the effect as treatment - control and after - before.

C. Under normal circumstances, a did model looks like this:



If γ is positive, our causal effect will be biased downwards due to the negative sign. If γ is negative, the estimator of the change will be biased upwards.